

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0971

Benton City 138 kV
October 30, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0971. The Interconnection Customer proposes a 60.4 MW battery energy storage facility interconnecting at the Benton City 138 kV substation in the Virginia area. The request was received on March 28, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 339 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,393,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WESTCOAST CELLUFIBRE to LEAVITT 138kV line	LEAVITT to WESTCOAST CELLUFIBRE 138kV Ckt 2	118.9	281	8.4
WESTCOAST CELLUFIBRE SUBSTATION to LEAVITT 138kV line	WESTCOAST CELLUFIBRE to LEAVITT SUBSTATION 138kV Cir. 1	118.9	281	8.4
KEOKUK to KENBRIDGE 138kV	MECKLENBURG POWER STA to KERR DAM 345kV L	118.1	320	5.2
KENBRIDGE to KLAMATH FALLS 138kV line	MECKLENBURG POWER STA to KERR DAM 345kV	117.6	326	5.2
KLAMATH FALLS to BAUXITE 138kV	KERR DAM to MECKLENBURG POWER STA 345kV line	117.0	333	5.2
ELKO to DANC LFGTE FACILITY 138kV	MIRA VISA to YRNEN 345kV	107.5	228	9.2
BRIMLEY to HNEYVILLE 138kV L	KERR DAM to MECKLENBURG POWER STATION 345kV	104.2	433	7.9
DAVIS to HOOD MESA 138kV	THALIA to LYNNHAVEN 138kV Line	102.5	241	9.2
E LONGMEADOW to PARSONS TAP 138kV	MCCARTNEY to YARNEN 345kV	102.2	344	8.5
OMNI-HACKBERRY to NORWICH 500kV	TREGO to BEAVER 500kV	100.2	3,369	7.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WESTCOAST CELLUFIBRE to LEAVITT 138kV #1	Rebuild line with higher-capacity conductor	\$1,798,000	2% (\$43,000)
WESTCOAST CELLUFIBRE to LEAVITT 138kV Line 2	Rebuild line with higher-capacity conductor	\$1,368,000	2% (\$32,000)
KEOKUK to KENBRIDGE 138kV	Rebuild line with higher-capacity conductor	\$9,455,000	4% (\$385,000)
KENBRIDGE to KLAMATH FALLS 138kV line	Rebuild line with higher-capacity conductor	\$12,290,000	4% (\$500,000)
BAUXITE to KLAMATH FALLS 138kV	Rebuild line with higher-capacity conductor	\$4,688,000	4% (\$191,000)
ELKO SUBSTATION to DANC LFGTE FACILITY 138kV Line	Reconductor line	\$3,154,000	5% (\$155,000)
BRIMLEY to HNEYVILLE 138kV L	Reconductor line	\$6,823,000	4% (\$269,000)
HOOD MESA SUBSTATION to DAVIS 138kV	Reconductor line	\$2,479,000	8% (\$193,000)
EAST LONGMEADOW to PARSONS TAP 138kV	Reconductor line	\$7,632,000	4% (\$312,000)
OMNI-HACKBERRY to NORWICH 500kV	Reconductor line	\$35,606,000	1% (\$313,000)
	Total allocated to project		\$2,393,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.